

iPhone OS

iOS

What is an Operating System

- ❑ Software that supports a computer's basic functions, such as scheduling tasks, executing applications, and controlling peripherals.
- ❑ Most important system software in computer system
- ❑ NOTE: User cannot run an application program on computer without OS

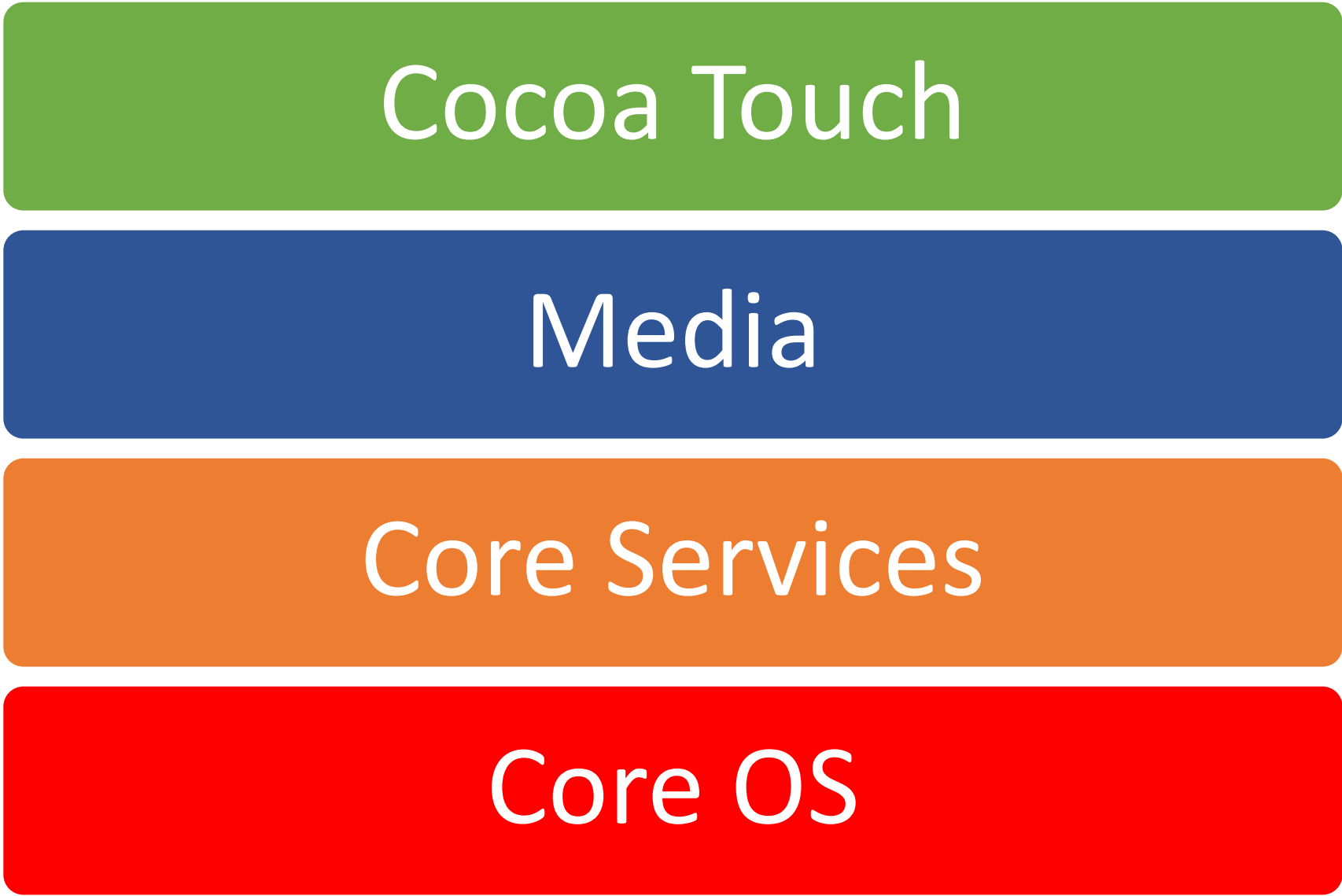
What is iOS ?



What is iOS ?

- ❑ iOS (formerly iPhone OS) is Apple's mobile operating system.
- ❑ Apple does not license I for installation on non-Apple hardware (distributed exclusively for Apple hardware)
- ❑ OS is derived from Mac OS X.
- ❑ Originally developed for the iPhone, but it's extended now to support other Apple devices such as the iPod Touch, iPad, and Apple TV.

What is iOS ? - Architecture



A diagram showing the four layers of iOS architecture as stacked horizontal bars. From top to bottom, the layers are: Cocoa Touch (green), Media (dark blue), Core Services (orange), and Core OS (red). Each layer is a rounded rectangle with its name centered in white text.

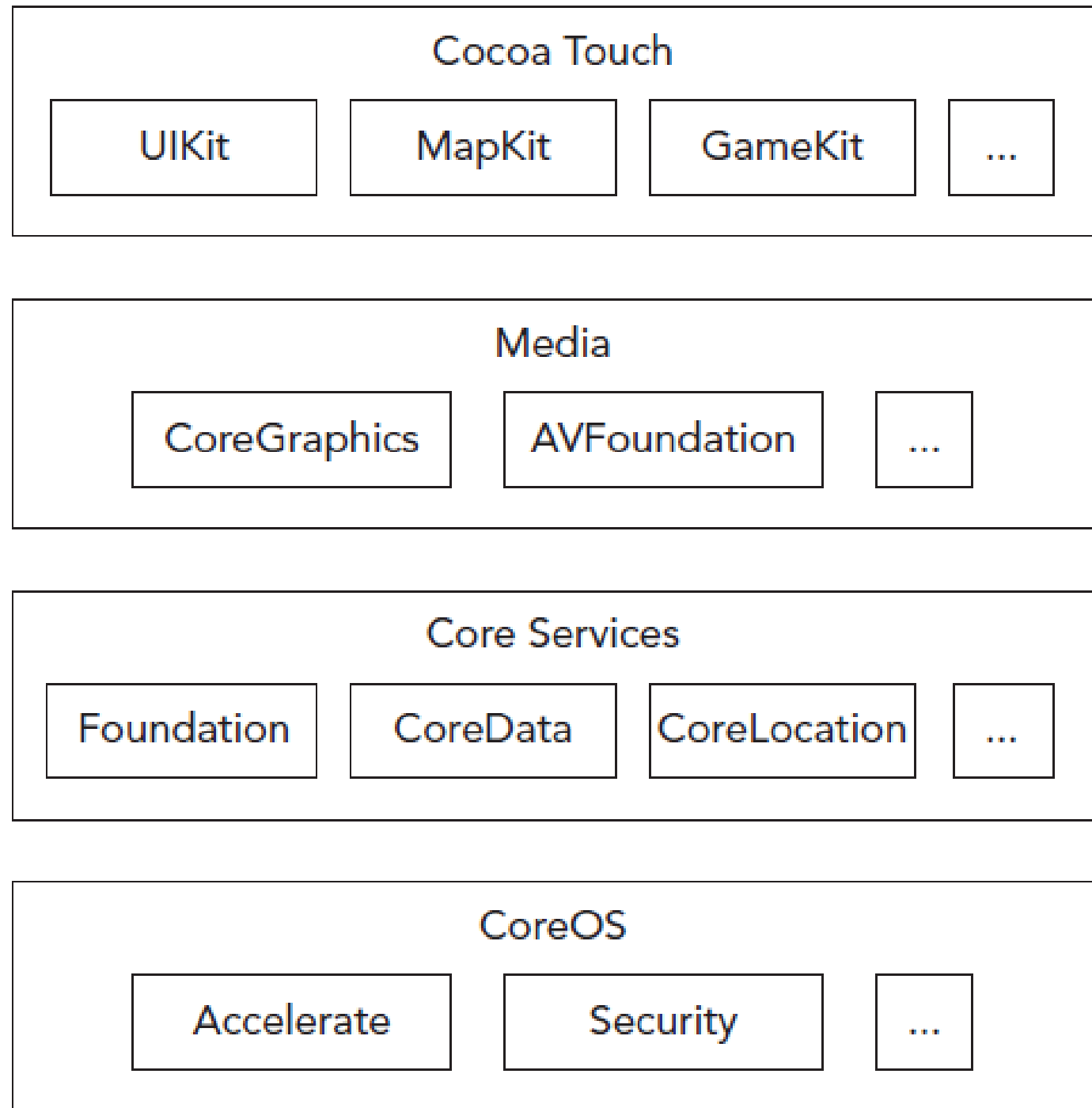
Cocoa Touch

Media

Core Services

Core OS

IOS Framework
Layers



Architecture - Cocoa Touch Layer

- ❑ The Cocoa Touch layer contains key frameworks for building iOS apps.
- ❑ These frameworks define the appearance of your app
- ❑ Include high-level features and Cocoa Touch Frameworks

Cocoa Touch Layer – High-level features

AirDrop

Text Kit

UIKit
Dynamics

Multitasking

Auto Layout

Storyboards

UI State
Preservation

Apple Push
Notifications

Local
Notifications

Gesture
Recognizers

Standard
System View
Controller

Cocoa Touch Frameworks

❑ Services Cocoa Touch Layer offers.

❑ Features:

❑ Address Book UI Framework

❑ Event Kit UI Framework

❑ Game Kit Framework

❑ iAd Framework

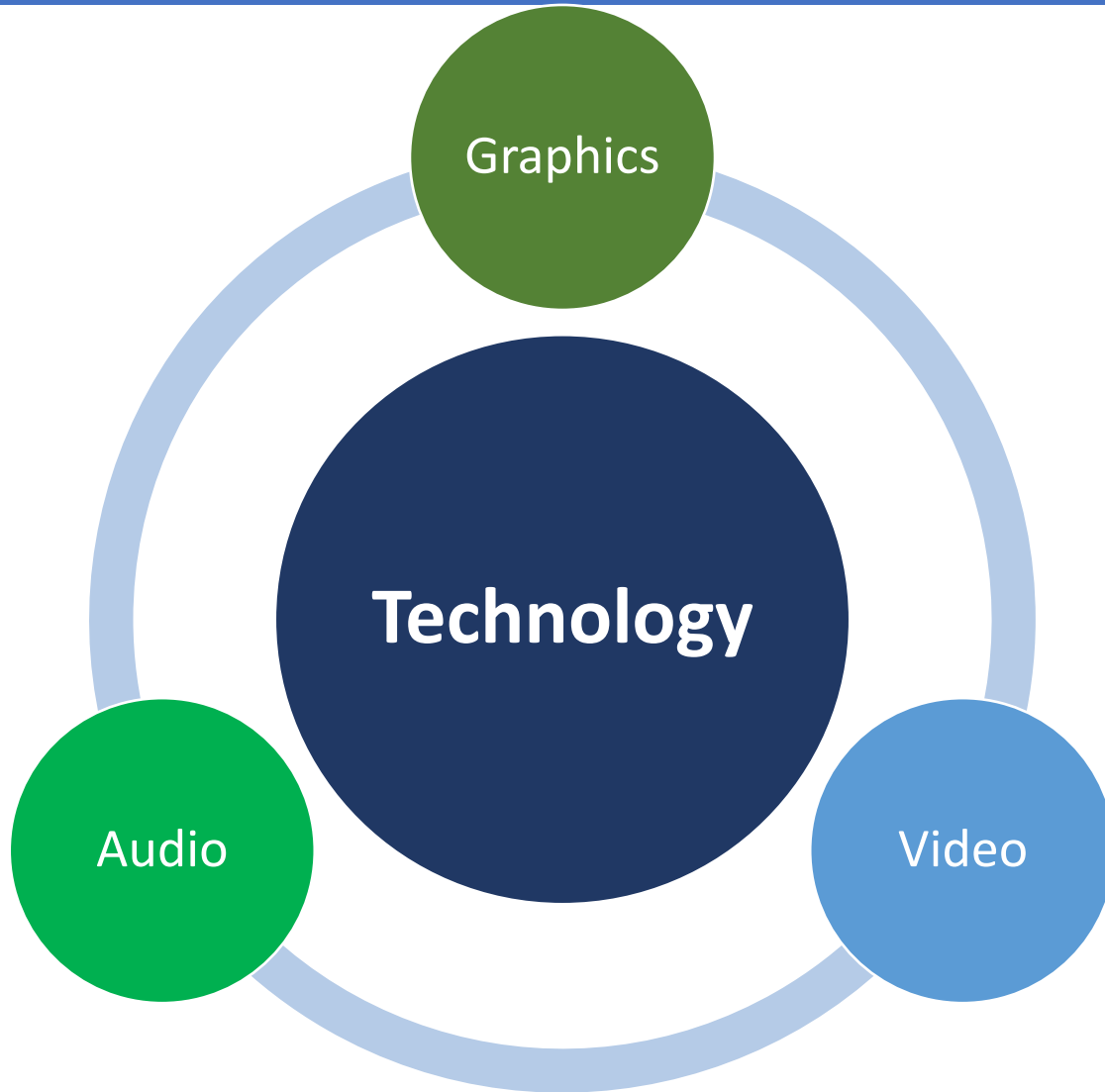
❑ Map Kit Framework

❑ Message UI Framework

❑ Twitter Framework

❑ UIKit Framework

Media Layer



Media Layer

- Core Graphics
 - 2D Vector and Raster Graphics
- Core Animation
 - View Animation
- Core Image (Objective-C)
 - Image and Video Manipulation, Filters
- OpenGL ES and GLKit
 - Hardware-accelerated 3D Graphics
- Core Text
 - Text Layout and Rendering
- Image I/O
 - Reading and Writing Images
- Assets Library
 - Access to User's Photos and Videos



Media Layer

- Media Player Framework (Objective-C)
 - Access to iTunes Library and Simple Playback
- AV Foundation Frameworks (Objective-C)
 - Audio and Video Capture and Playback
- OpenAL
 - Positional Audio
- Core Audio Framework
 - Advanced Audio Playback
- Core Video Framework
 - Advanced Video Playback
- AirPlay (Objective-C)
 - Stream Audio and Video to Other Devices



Core Layer

☐ High-Level Features

☐ Peer-to-Peer Services

☐ iCloud Storage

☐ Automatic Reference Counting

☐ Data Protection

☐ File-Sharing Support

☐ Grand Central Dispatch

☐ In-App Purchase

☐ SQLite

☐ XML Support

Core Layer

☐ Core Services Frameworks

- ☐ Accounts Framework
- ☐ Address Book Framework
- ☐ CFNetwork Framework
- ☐ Core Data Framework
- ☐ Core Foundation Framework
- ☐ Core Location Framework
- ☐ Core Media Framework
- ☐ Core Motion Framework
- ☐ Core Telephony Framework
- ☐ Social Framework

Core OS Layer

Core OS Layer

Accelerate Framework

Core Bluetooth Framework

External Accessory Framework

Generic Security Framework

Security

System

64-bit Support

Features

- ☐ Home Screen
- ☐ Included Applications
- ☐ Multitasking
- ☐ Siri
- ☐ Game Center

Jailbreaking

- ❑ The process of removing limitations on iOS, Apple's operating system, on devices running it through the use of software and hardware exploits.
- ❑ Target: iPhone, iPod touch, iPad and second-generation Apple TV.
- ❑ Why is “Jailbreak” ?
 - ❑ allowing the installation of software that is unavailable through the official [Apple App Store](#).

Jailbreaking

- ❑ A jailbroken running iOS can still use the App Store, iTunes, and other normal functions.
- ❑ Potentials of Security, Privacy and Stability